Lab 3: Component Modelling & Architectural Pattern Selection

Digital Campus Library Reservation Gateway — Architectural Specification & Component Model

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1. Scenario Review & Requirements Engineering

System: Digital Campus Library Reservation Gateway (Domain: Campus & Academic Operations). The system automates the circulation lifecycle of high-demand textbooks across university libraries. It eliminates manual waitlists, shelf idle latency, and tracking overhead via ISBN catalog search, automated FIFO hold queues, multi-channel pickup alerts (48h window), and dynamic overdue fine enforcement (Rs 10/day post-grace period).

1.1 Functional Requirements Specification (FR-001 to FR-005)

ID & Priority	Requirement Description	Acceptance Criteria (Pass / Fail)	Rationale & Business Value
FR-001 High (Booking)	**Catalog Search & Hold:** Allow student members to search ISBN catalog and place hold requests on loaned copies to enter dynamic FIFO reservation queue.	**Pass:** Eligible student assigned next sequential integer in FIFO queue; reservation token generated. **Fail:** Student holding >=3 active holds or fines >Rs 50 places hold.	Enforces fair, transparent, strictly automated textbook allocation.
FR-002 High (Queue)	**Dynamic FIFO Reallocation:** Automatically advance waitlist and allocate available copy to #1 queued student on return check-in or hold expiry.	**Pass:** Book flagged 'Reserved-Ready', locked for #1 student, 48h pickup window initiated. **Fail:** Copy allocated out-of-order or remains idle.	Minimizes shelf idle latency; eliminates manual librarian matching.
FR-003 Medium (Alert)	**Multi-Channel Notification:** Dispatch real-time SMS and email alerts when held book is ready, approaching expiration (24h warning), or overdue.	**Pass:** Alert dispatched within 60s of state change with shelf code and deadline. **Fail:** Dispatch fails without retry log.	Maximizes pickup compliance; accelerates circulation turnover.
FR-004 High (Finance)	**Overdue Fine Calculation & Settlement:** Automatically accrue daily overdue fines (Rs 10/day post-grace period), restrict delinquent accounts, and log settlement upon payment.	**Pass:** Fine accrues at 00:00 hrs; account unblocks immediately upon full payment. **Fail:** Delinquent student places hold or borrows.	Incentivizes timely returns; maintains immutable institutional audit ledger.
FR-005 Medium (Admin)	**Catalog & Queue Override:** Allow Head Librarian to update ISBN records, view live waitlists, manually override queue priority with audit logging, and export analytics.	**Pass:** Inventory modified, queue reordered with recorded justification, CSV/PDF exported. **Fail:** Priority override executed without logged reason.	Provides operational flexibility for emergencies while preserving audit traceability.

1.2 Non-Functional Requirements Specification (NFR-001 & NFR-002)

ID & Type	Requirement Description	Acceptance Criteria (Pass / Fail)	Technical Rationale
NFR-001 Performance & Scalability	Catalog search queries shall return paginated results for 50,000+ titles in <150 ms under concurrent load of up to 1,000 active student sessions.	**Pass:** 95th percentile latency <=150 ms, throughput >=500 req/sec under peak load. **Fail:** Latency >250 ms or dropped connections.	Prevents campus network saturation during peak semester rush; guarantees smooth UX.
NFR-002 Security & Integrity	The system shall enforce Role-Based Access Control (RBAC), OAuth 2.0/JWT token authentication, and encrypted audit logs for overrides and financial transactions.	**Pass:** All endpoints validate cryptographic signatures; zero unauthorized privilege escalation; audit checks pass. **Fail:** Override or fine change without valid admin role.	Guarantees reservation queue integrity against tampering; secures financial ledgers.

2. Architectural Style Analysis & Evaluation

In accordance with Lab 3 objectives, three architectural styles were evaluated against the requirements of the Digital Campus Library Reservation Gateway:

Architectural Style	Key Characteristics & Structure	Advantages (Pros)	Disadvantages (Cons) & Scenario Fit
1. Layered Architecture (Selected)	Organized in distinct horizontal tiers: Presentation, Business Logic, and Data Persistence. Components interact strictly via defined provided/required interfaces.	• Clear separation of concerns. • Simplified data consistency & ACID compliance. • Easy to develop, test, and maintain. • High throughput via in-memory caching.	• Tight coupling between adjacent layers. **Fit: Optimal.** Directly supports atomic FIFO queue locking and RBAC security perimeter without distributed complexity.
2. Microservices Architecture	System decomposed into small, fine-grained, independently deployable services communicating over network via HTTP REST or gRPC.	• Independent scaling of services. • Fault isolation. • Technology heterogeneity. • Independent deployments.	• Distributed transaction complexity (SAGA/2PC). • Network latency overhead (>50ms network hops). **Fit: Sub-optimal.** Unnecessary overhead for single campus deployment.
3. Client-Server Architecture	Two-tier system with multiple client applications connecting directly to a centralized server hosting business logic and database.	• Centralized data management. • Simple initial deployment model. • Straightforward client connectivity.	• Single point of failure. • Scalability bottleneck under concurrent load. • Poor modularity. **Fit: Poor.** Violates NFR-001 concurrency targets.

3. Architecture Selection & Written Justification

Architecture Selection: "We chose Layered Architecture for the Digital Campus Library Reservation Gateway System."

3.1 Architectural Choice

We selected a **3-Tier Layered Architecture** structured into: (1) **Presentation Layer** (User Interface Component handling Web & Mobile student/librarian portals and SSO validation); (2) **Business Logic Layer** (Catalog Search Component, Reservation & Queue Manager Component, Payment & Fine Service Component, and Notification Service Component); and (3) **Data Layer** (Database Component managing catalog tables, active loans, FIFO queue records, and immutable audit logs).

3.2 Two Specific Scenario-Related Reasons for Selection

Reason 1: Dynamic FIFO Queue Integrity & Atomic State Transitions (FR-001 & FR-002): The primary domain challenge is guaranteeing strict FIFO waitlist ordering upon book return check-in, atomically locking the physical copy for the #1 student, and initiating the 48-hour pickup window. Collocating queue management logic with an ACID relational data tier allows row-level locks on book reservation records during return check-in. This completely prevents race conditions, double-allocations, and out-of-order assignments without requiring complex distributed consensus protocols.

Reason 2: Clean Separation of Domain Rules for Delinquency & Inventory Governance (FR-004 & FR-005): The business layer cleanly encapsulates institutional policies—such as automated overdue fine calculation (Rs 10/day post-grace period), delinquent hold blocks (>Rs 50 balance or >3 active holds), and librarian priority overrides with audit logging. By decoupling these rules from the presentation tier and database schema, academic circulation policies can be updated without modifying client interfaces or storage tables.

3.3 Security Advantage

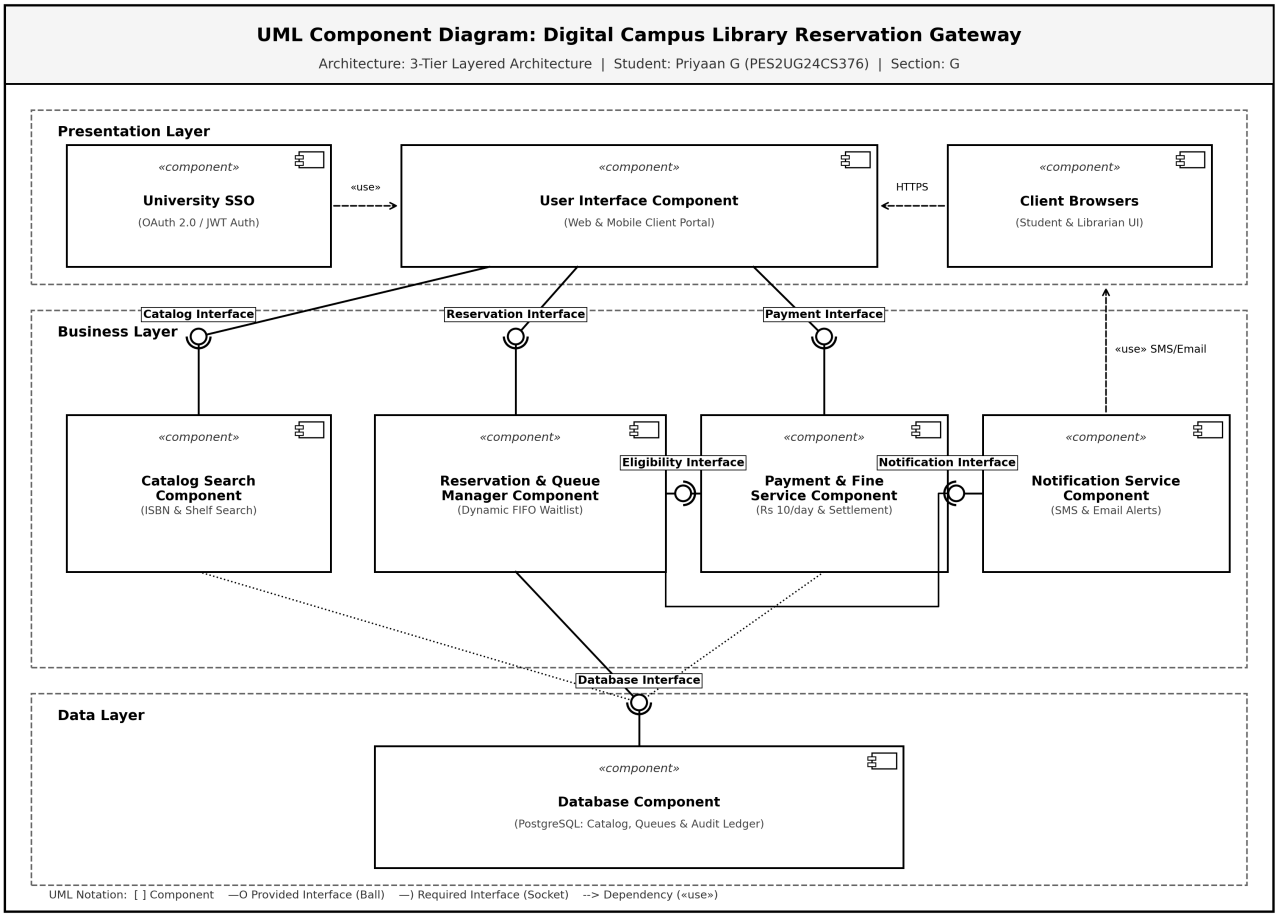
Perimeter Authentication & Isolated Persistence Boundary (NFR-002): Layered architecture enforces a strict security perimeter. Client applications and student browsers communicate solely with the Presentation Layer via HTTPS, which validates University SSO OAuth 2.0 / JWT tokens before relaying requests. Direct external access to the database is prohibited; all data interactions are mediated by the Business Layer's Role-Based Access Control (RBAC) engine. This prevents SQL injection, guarantees zero unauthorized privilege escalation, and preserves an immutable audit trail for financial settlements and queue overrides.

3.4 Performance Benefit

Sub-150 ms Query Latency via Collocated Caching & Zero Network Hop Overhead (NFR-001): Under peak semester rush (1,000+ concurrent student sessions and 50,000+ titles), layered architecture enables tight coupling with an in-memory catalog cache directly accessible by the business tier. This avoids inter-service network hops, serialization latency, and network partitioning bottlenecks typical of microservices, ensuring 95th-percentile catalog search latency remains reliably under 150 ms while sustaining over 500 req/sec.

4. UML Component Diagram & Interface Architecture

The UML Component Diagram below follows UML 2.5 standards and the visual conventions of the Lab 3 handout (simple black-and-white notation, horizontal layer grouping, standard component boxes, provided ball interfaces, and required socket interfaces).



4.1 Component & Interface Traceability Matrix

Component Name	Layer	Provided Interfaces (Ball)	Required Interfaces (Socket)	Mapped Requirements
User Interface Component	Presentation	• None (Client UI Host)	• Catalog Interface • Reservation Interface • Payment Interface	FR-001, FR-004, FR-005, NFR-002
Catalog Search Component	Business	• **Catalog Interface:** searchTitles(isbn, query) checkAvailability(copyId)	• Database Interface	FR-001, NFR-001 (<150ms)
Reservation & Queue Manager	Business	• **Reservation Interface:** placeHold(studentId, isbn) cancelHold(holdId) reorderQueue(adminId)	• Eligibility Interface • Notification Interface • Database Interface	FR-001, FR-002, FR-005
Payment & Fine Service	Business	• **Payment Interface:** settleFine() • **Eligibility Interface:** isEligible()	• Database Interface	FR-004 (Rs 10/day, >Rs 50 block)
Notification Service	Business	• **Notification Interface:** dispatchPickupAlert() dispatchExpiryNotice()	• External SMS/Email Provider («use» REST/SMTP)	FR-003 (SMS/Email within 60s)
Database Component	Data	• **Database Interface:** queryCatalog() • atomicQueueLock() • appendAuditLog()	• Storage System (PostgreSQL)	NFR-002 (ACID & Audit Integrity)